Heterogeneous SoC Architecture: PS/PL Acceleration on Digilent Cora Z7-10

Xilinx Zynq-7010 (XC7Z010-1CLG400C) | ARM Cortex-A9 @ 667 MHz + True Dual-Port BRAM Decoupled LZSS & AES-128-CTR Accelerators

DELT UFMG - Sistemas em Silício
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EXTERNAL STIMULUS (ESP32-WROOM)

Bare-Metal Link

Core Architecture
Xtensa 32-bit Dual-Core LX6 @ 240 MHz

Firmware Stack
ESP-IDF v6.1 FreeRTOS Bare-Metal Injector

Physical Interface
SPI Master (Mode 0, MSB-First, Up to 25 MHz)

Frame Framing
Preamble 0xA55A | SeqID [31:0] | Reserved [15:0]

Payload Spectrum
Dynamic Cycling: 64 B, 256 B, 512 B, 1024 B

Integrity & Guard
CRC-16 CCITT (0x1021) + EOF Delimiter 0x5A

Autonomous Trigger
Active-Low Handshake on GPIO 32 (Ext Int)

PHYSICAL HEADER J7 / PMOD JA PINOUT

- **SPI SCLK : ESP32 GPIO 18 -> Cora Z7 J7.3 (Pin H15)**
- **SPI MOSI : ESP32 GPIO 23 -> Cora Z7 J7.4 (Pin T12)**
- **SPI MISO : ESP32 GPIO 19 <- Cora Z7 J7.1 (Pin W15)**
- **SPI SS_N : ESP32 GPIO 5 -> Cora Z7 J7.5 (Pin F16)**
- **BURST TRIG : ESP32 GPIO 32 <- Cora Z7 PMOD JA1 (Pin Y18)**
- **GROUND : Common GND <-> Cora Z7 Header GND**

BUS ARCHITECTURE & COLOR KEY

- Physical Off-Chip SPI & Trigger (Header J7 / PMOD JA)
- AXI4-Lite Control Bus (CSRs @ 0x41E0/43C0/43C1)
- AXI4 BRAM Memory Bus (Port A @ 0x4000/4001)
- Native Synchronous BRAM Port B (RTL Engine Datapath)
- ARM Cortex-A9 Processing System (PS @ 667 MHz)
- Artix-7 Programmable Logic Fabric (PL @ 100 MHz)

TOTAL PL UTILIZATION (XC7Z010 Budget):

LUTs (Logic) : 8,856 / 17,600 (50.3% Utilization)
Registers (FFs) : 12,047 / 35,200 (34.2% Utilization)
Block RAM (RAMB36) : 9.0 / 60 (15.0% Utilization)
DSP48E1 Slices : 0 / 80 (0.0% (Pure Logic))
Worst Negative Slack: +1.242 ns (Timing MET (100 MHz))

PROCESSING SYSTEM 7 (PS - Cortex-A9 @ 667 MHz)

Bare-Metal BSP / Standalone OS

ARM Cortex-A9 MPCore (Core 0)

- 667 MHz Peak Frequency | VFPv3-D32 + NEON SIMD
- 32 KB L1 Instruction + 32 KB L1 Data Caches
- ARM PMU (CP15 Cycle Counter, 29 Cyc Overhead)
- Golden Software Pipeline (Heatshrink + NIST AES-CTR)

DDR3 Memory Controller (512 MB)

- 16-bit Bus @ 533 MHz (1066 Mbps DDR3-1066)
- Staging Ingress Telemetry Buffers (Raw / Cipher)
- High-Precision Timing Calibration & Logging Arrays
- Zero Dynamic Allocation (Deterministic MISRA C)

M_AXI_GP0 (50 MHz)

Physical SPI Link: Header J7 (H15, T12, W15, F16)

PMOD JA1 (Y18) -> ESP32 GPIO 32 Trigger

PS Control Peripherals & Interconnect Anchors:

- DEVCFG: Address 0xF800700C (Bit 2 PCFG_DONE Interlock - Prevents CPU Bus Hangs)
- EMIO GPIO: 1-Bit Slice routed to PMOD JA1 (Pin Y18) for Autonomous 30 ms ESP32 Trigger
- M_AXI_GP0: 32-bit AXI Master Interface driving PL Interconnect @ 50 MHz (Synchronous)

AXI Address Map Master Table (M_AXI_GP0):

```
0x41E00000 : AXI Quad SPI 0 (Ingress FIFO) | 0x43C00000 : LZSS Accel CSRs
0x43C10000 : AES-128-CTR Accel CSRs        | 0x40000000 : LZSS In BRAM (8 KB)
0x40002000 : LZSS Out BRAM (8 KB)          | 0x40010000 : AES In BRAM (8 KB) | 0x40012000 : AES Out BRAM
```

PROGRAMMABLE LOGIC (PL - Artix-7 Fabric)

FPGA Accelerators & BRAM Memory Hierarchy

AXI Interconnect: ps7_0_axi_periph (Crossbar + Clock Converters 50M -> 100M)

Demultiplexes M_AXI_GP0 into 7 Master Channels: 1x SPI IP, 2x Accel CSRs, 4x AXI BRAM Controllers

AXI4-Lite: 0x41E00000

Unified Ingress: axi_quad_spi_0 (Base: 0x41E00000)

- Slave Mode 0 (CPOL=0, CPHA=0) with 256-Byte Hardware FIFO | Status Register SPISR = 0x25
- Physical Pins: SCLK (H15), MOSI (T12), MISO (W15), SS (F16) connected to Cora Z7 Shield Header J7
- Firmware Ingestion Service drains FIFO over AXI4-Lite into DDR3 staging buffers before hardware offload
- Zero Frame Corruption: 100% CRC-16 CCITT parity verified across 64 B to 1024 B payloads

AXI BRAM Ctrl: 0x40000000 | AXI4-Lite CSRs: 0x43C00000 | AXI BRAM Ctrl: 0x40002000

HARDWARE ACCELERATOR 1: lzss_accel (Sliding-Window Dictionary Compressor)

Base CSR: 0x43C00000 | FCLK1 = 100 MHz | Window = 256 Bytes | Lookahead = 16 Bytes | Heatshrink Bit-Exact Format

Input BRAM (bmg_lzss_in)

- Capacity: 8 KB (2048 x 32-bit words)
- Port A: AXI BRAM Controller @ 0x40000000 (PS writes raw telemetry frames via MMIO)
- Port B: Native Synchronous Accelerator Bus (clk, addr[12:0], din, dout[31:0], we)
- True Dual-Port Independent Clocks & BRAM

Native Port B

lzss_accel RTL Core (VHDL-2008)

- FSM: IDLE -> READ_IN -> SEARCH -> EMIT -> WRITE
- Sliding Window: 256 B circular buffer search
- Variable-length bitstream packer & flag byte
- Local 32-bit PL Hardware Cycle Counter
- Resources: 2,840 LUTs | 1,420 FFs | 2 BRAMs
- Empirical PL Latency: 269,675 cyc (1024 B)

Native Port B

Output BRAM (bmg_lzss_out)

- Capacity: 8 KB (2048 x 32-bit words)
- Port B: Native Synchronous Accelerator Bus (Accelerator writes compressed tokens)
- Port A: AXI BRAM Controller @ 0x40002000 (PS reads back compressed bitstream)
- Comp. Ratio: 54.9% (1024 B -> 562 B)

MICROARCHITECTURAL BOTTLENECK ANALYSIS (RISK R9):

Sequential BRAM Port B linear search @ 100 MHz requires up to 768 cyc/byte. Cortex-A9 @ 667 MHz with L1/L2 cache completes faster (0.596 ms vs 0.951 ms).

AXI BRAM Ctrl: 0x40010000 | AXI4-Lite CSRs: 0x43C10000 | AXI BRAM Ctrl: 0x40012000

HARDWARE ACCELERATOR 2: aes_accel (NIST SP 800-38A AES-128-CTR Cryptographic Engine)

Base CSR: 0x43C10000 | FCLK1 = 100 MHz | 10-Round Iterative Pipeline | Keystream XOR | 11 Cycles / 16-Byte Block

Input BRAM (bmg_aes_in)

- Capacity: 8 KB (2048 x 32-bit words)
- Port A: AXI BRAM Controller @ 0x40010000 (PS writes compressed payload for encryption)
- Port B: Native Synchronous Accelerator Bus (Acc reads 128-bit blocks into cipher)
- Dual-port isolation prevents bus contention

Native Port B

aes_accel RTL Core (VHDL-2008)

- SubBytes (S-Box) -> ShiftRows -> MixColumns
- 128-bit Round Key Expansion on-the-fly
- Big-Endian Counter Increment & Keystream XOR
- Local 32-bit PL Hardware Cycle Counter
- Resources: 3,410 LUTs | 2,150 FFs | 2 BRAMs
- Measured PL Cycles: 829 cyc (1024 B payload)

Native Port B

Output BRAM (bmg_aes_out)

- Capacity: 8 KB (2048 x 32-bit words)
- Port B: Native Synchronous Accelerator Bus (Acc writes ciphertext blocks in real-time)
- Port A: AXI BRAM Controller @ 0x40012000 (PS retrieves encrypted frame for staging/ACK)
- 100% NIST SP 800-38A Golden Parity Verified

PEAK HETEROGENEOUS ACCELERATION (CENÁRIO 3: SW LZSS + PL AES-CTR):

AES hardware offload eliminates 360,000 CPU cycles in pure SW! 256 B: 0.381 ms (vs 0.596 ms SW) -> 1.56x Net System Speedup (0.64 MB/s Throughput).

EMPIRICAL BENCHMARK CONSOLIDATION & SYSTEM METRICS

12 Target Measurements

Scenario	Config	Payload	Latency	CPU Cyc	PL Cyc	Speedup
Case 1 (SW/SW)	SW LZSS + SW AES	256 B	0.596 ms	397,343	0	1.00x (Ref)
Case 1 (SW/SW)	SW LZSS + SW AES	512 B	2.024 ms	1,349,888	0	1.00x (Ref)
Case 1 (SW/SW)	SW LZSS + SW AES	1024 B	4.846 ms	3,232,400	0	1.00x (Ref)
Case 2 (HW/SW)	PL LZSS + SW AES	256 B	0.951 ms	634,308	25,631	0.63x (R9)
Case 2 (HW/SW)	PL LZSS + SW AES	512 B	3.210 ms	2,141,194	107,539	0.63x (R9)
Case 2 (HW/SW)	PL LZSS + SW AES	1024 B	7.695 ms	5,132,354	269,675	0.63x (R9)
Case 3 (SW/HW)	SW LZSS + PL AES	256 B	0.381 ms	254,021	217	1.56x (Peak)
Case 3 (SW/HW)	SW LZSS + PL AES	512 B	1.580 ms	1,054,168	421	1.28x (Peak)
Case 3 (SW/HW)	SW LZSS + PL AES	1024 B	3.953 ms	2,636,750	829	1.23x (Peak)
Case 4 (HW/HW)	PL LZSS + PL AES	256 B	0.734 ms	489,777	25,848	0.81x
Case 4 (HW/HW)	PL LZSS + PL AES	512 B	2.767 ms	1,845,887	107,960	0.73x
Case 4 (HW/HW)	PL LZSS + PL AES	1024 B	6.794 ms	4,531,431	270,504	0.71x